Django Lab Demo

1. Create and open the project folder

Create a folder on the Desktop:

```
djangoproject
```

Open the folder in **VS Code**.

Then open:

Terminal → New Terminal

---

2. Create a virtual environment

In the VS Code terminal:

```
python3 -m venv venv
```

Activate it:

```
source venv/bin/activate
```

After activation, the terminal should show:

```
(venv)
```

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3. Install Django

Since Django was not available inside the new virtual environment, install it:

```
python -m pip install django
```

Check the Django version:

```
python -m django --version
```

The manual also uses a project-specific virtual environment and checks the installed Django version before continuing.

---

4. Create the Django project

Run:

```
django-admin startproject config .
```

Important

Do not forget the . at the end.

The project structure becomes:

```
djangoproject/
│
├── config/
├── venv/
└── manage.py
```

The . tells Django to create the project in the current folder.

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5. Create a Django application

Run:

```
python manage.py startapp pages
```

Now the structure becomes approximately:

```
djangoproject/
│
├── config/
│
├── pages/
│   ├── admin.py
│   ├── apps.py
│   ├── models.py
│   ├── tests.py
│   ├── views.py
│   └── migrations/
│
├── venv/
└── manage.py
```

---

6. Register the application

Open:

`config/settings.py`

Find:

```
INSTALLED_APPS = [
```

Add:

```
'pages',
```

So it becomes:

```
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'pages',
]
```

Save the file.

This is the registration step specified in the practice manual.

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7. Create the first View

Open:

`pages/views.py`

Replace the existing content with:

```
from django.http import HttpResponse

def home(request):
    return HttpResponse("Hello! This is my first Django page.")
```

Save the file.

A Django view is the Python function that receives the request and returns the response.

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8. Create the application's URL file

Inside the `pages` folder, create:

```
urls.py
```

Add:

```
from django.urls import path
from . import views

urlpatterns = [
    path('', views.home, name='home'),
]
```

Save the file.

---

9. Connect the app URL to the project URL

Open:

```
config/urls.py
```

Replace its contents with:

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('pages.urls')),
]
```

Save the file.

The request now follows this path:

```
Browser
   ↓
config/urls.py
   ↓
pages/urls.py
   ↓
views.py
   ↓
home()
   ↓
```

```
HttpResponse
   ↓
Browser
```

This is the request flow demonstrated in Practice Manual 1.

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10. Start the Django development server

In the VS Code terminal:

```
python manage.py runserver
```

You should see:

```
Starting development server at http://127.0.0.1:8000/
```

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11. Open the website

Open your browser and enter:

```
http://127.0.0.1:8000/
```

You should see:

```
Hello! This is my first Django page.
```

First Django webpage successfully created.

The manual uses the same URL and checkpoint for the first working page.

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12. Demonstrate a change

Now go back to:

```
pages/views.py
```

Change:

```
return HttpResponse("Hello! This is my first Django page.")
```

to:

```
return HttpResponse("Welcome to Django Web Development!")
```

Save the file.

Refresh the browser.

You should now see:

```
Welcome to Django Web Development!
```

This is a good point to tell students:

"The browser is displaying the response generated by our Python view."

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13. Final structure

At the end of this part, students should have:

```
djangoproject/
│
├── config/
│   ├── __init__.py
│   ├── settings.py
│   ├── urls.py
│   ├── asgi.py
│   └── wsgi.py
│
├── pages/
│   ├── migrations/
│   ├── __init__.py
│   ├── admin.py
│   ├── apps.py
│   ├── models.py
│   ├── tests.py
│   ├── urls.py
│   └── views.py
│
├── venv/
│
└── manage.py
```

Commands students need to remember

```
python3 -m venv venv
source venv/bin/activate
python -m pip install django
```

```
python -m django --version
django-admin startproject config .
python manage.py startapp pages
python manage.py runserver
```

One-slide summary

You can put this on the projector before starting:

```
DJANGO FIRST WEB PAGE — LAB DEMO

1. Create project folder
2. Open folder in VS Code
3. Open Terminal
4. Create virtual environment
5. Activate virtual environment
6. Install Django
7. Create Django project
8. Create Django app
9. Register app in settings.py
10. Create View
11. Create app URL
12. Connect app URL to project URL
13. Run Django server
14. Open http://127.0.0.1:8000/
```

The key concept

```
URL → VIEW → RESPONSE
```

And in our implementation:

```
config/urls.py
      ↓
pages/urls.py
      ↓
pages/views.py
      ↓
HttpResponse
      ↓
Browser
```